

Phase	Primary Roles Responsible	What They Do
Problem Definition	Business Analyst, Product Manager, Lead Data Scientist	Translate high-level business goals into measurable project objectives and determine if the project is feasible.
Data Collection	Data Engineer	Build data pipelines, query databases, connect APIs, and ensure the raw data arrives reliably and securely.
Data Preparation	Data Engineer, Data Scientist	Clean messy records, structure tables, handle missing entries, and transform columns into formats suitable for analysis.
EDA (Exploratory Data Analysis)	Data Analyst, Data Scientist	Create initial charts, spot patterns, identify outliers, and check assumptions to understand what the data actually says.
Modelling	Data Scientist, Machine Learning (ML) Engineer	Select algorithms, train models on the prepared data, tune settings, and optimize for predictive accuracy.
Evaluation	Data Scientist, Business Stakeholder, Domain Expert	Verify the model meets real-world performance targets, check for bias or unintended errors, and ensure leadership

		understands the results.
Deployment	Machine Learning (ML) Engineer, MLOps Engineer, Software Engineer	Package the model, build APIs, and integrate it into live applications, websites, or automated workflows.
Monitoring and Maintenance	MLOps Engineer, Site Reliability Engineer (SRE), Data Scientist	Track uptime, watch for shifts in data patterns (drift), fix live bugs, and schedule model retraining.